

Predicting Electricity Prices Using Time Series Analysis

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PROBLEM STATEMENT AND MOTIVATION

In electrical markets, energy price forecasting is essential because it helps producers, suppliers, and consumers make well-informed financial and operational decisions. Market players may optimize production plans, reduce risk, and enhance bidding strategies in both day-ahead and real-time markets with the help of accurate price forecasts. Creating accurate forecasting models is still quite difficult because of the inherent volatility of energy prices, which are impacted by a number of variables, including weather, fuel prices, renewable energy sources, and changes in demand. Despite this, modern statistical analysis proves to be up to the challenge. Enabled by vast swaths of market data, we can now produce accurate results with ostensibly simple models.

With that in mind, we wanted to replicate and even improve on results from industry-standard, widely used methods. Specifically, **the central objective of this project is to explore whether machine learning techniques, combined with ensemble and stacking methods can improve short-term electricity price forecasting relative to individual statistical models and the official TSO forecasts.** Baseline models such as seasonal naive forecasting, as well as traditional time series approaches like ARIMA and SARIMAX, are employed as first-level predictors. These individual model forecasts are then combined using XGBoost as a meta-learner, under the hypothesis that each model captures different underlying patterns in the data, and their combination can lead to more accurate and robust predictions.

INTRODUCTION AND DESCRIPTION OF DATA

This project utilizes a dataset sourced from Kaggle (procured by Nicholas Shaw), containing four years of hourly electricity consumption, generation, pricing, and weather data for Spain. The consumption and generation data were retrieved from ENTSOE, the European Network of Transmission System Operators for Electricity, while settlement prices were obtained from Red Eléctrica de España (REE), Spain's national TSO (Transmission System Operator). The dataset also includes weather data from the Open Weather API for Spain's five largest cities. A unique aspect of this dataset is its inclusion of both actual prices and the official day-ahead price forecasts provided by the TSO, allowing for direct benchmarking of forecasting models against industry-standard predictions.

Data Description

Predictors, Target Variable, Summary Statistics

The dataset includes a total of 27 predictors, one of which is time (observations are recorded hourly), with the rest being “exogenous”: external factors that influence the target variable being predicted. Exogenous variables tend to be so-called “generation” statistics, which detail reported amounts of electricity production by different energy sources, measured in megawatts. Some generation statistics include electricity generated by biomass sources, fossil fuel generated by natural gas, and geothermal energy generation. Other notable predictors include total load, which is the measured electrical demand at each hour, and total load forecast, the official forecasted demand one day ahead. The target variable, “price_actual”, is the actual price of electricity in Euros per megawatt. Our dataset includes 35,064 total observations.

EXPLORATORY DATA ANALYSIS

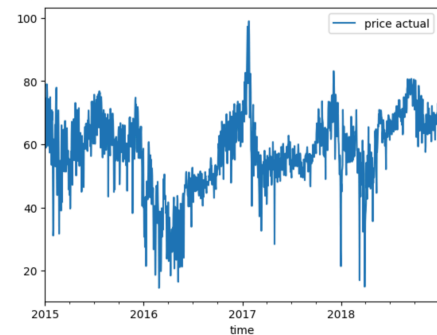
EDA is a crucial aspect of time series analysis; through visualization, EDA helps identify trends and seasonality in the data, while also revealing potential issues like assumption violations, missing

values, and variance. For time series specifically, understanding these aspects of the data is absolutely necessary for our models to work properly, as EDA can help inform whether differencing is needed to achieve stationarity or whether seasonal components should be incorporated. Additionally, by including feature selection in our EDA, we ensure that our forecasts will be as robust as possible and increase the chance of higher accuracy models later on.

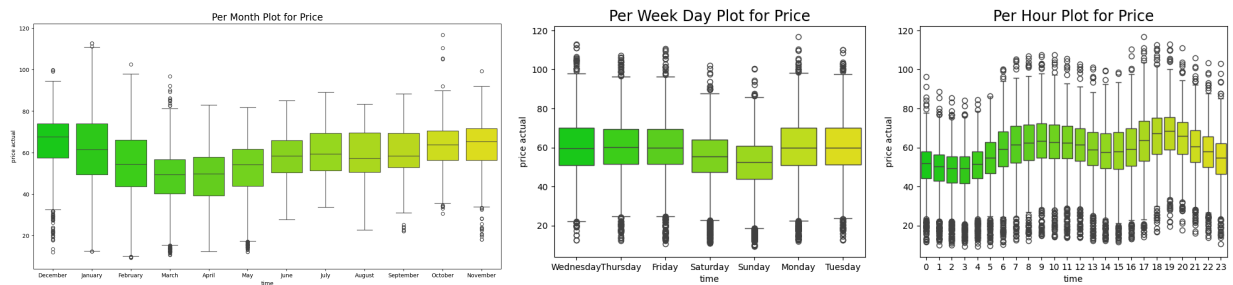
Target Variable Analysis

Analyzing the distribution and seasonality of the response variable, hourly electricity price.

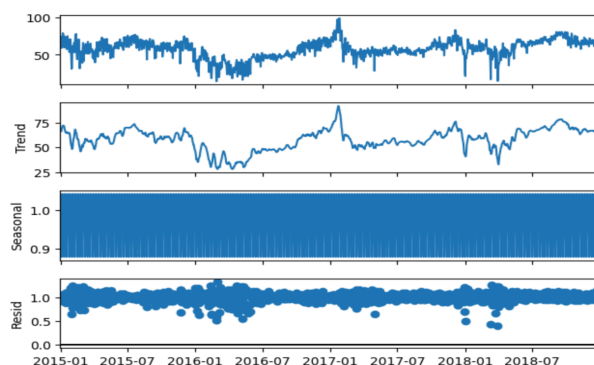
This line plot to the right displays the changes in energy prices from 2015-2018. In our problem, this graph provides insight into the more long term volatility and seasonal changes seen in energy prices, revealing noticeable trends. One such trend is a relatively sharp decrease in prices towards the end of 2015 to early 2016, followed by a more steady increase and eventual spike in 2017. Identifying these spikes and drops within our data will better allow us to predict future prices as we can further understand how price fluctuations and changes have appeared in past years.



To get a better look at trends and patterns, our initial approach focused on visualizing energy prices on a monthly basis. Thus, we began by plotting our Actual Prices variable for each month, day, and hour per day using box plots. This allowed us to examine the distribution of our data, as well as identify any clear trends in seasonality..



The first box plot displays the distribution of electricity prices for each. Each box plot represents the interquartile range of prices for each month, with the line inside the box indicating the median price. The whiskers represent the range and the circles outside the whiskers indicate the outliers. The plot reveals that the electricity prices tend to fluctuate more during certain months, with higher prices showing in the winter from December to February and summer from July to August. This indicates that there could potentially be seasonal patterns likely due to increased energy consumption during periods of more heating and cooling. Similarly we looked at both Day of week and hour both in which had less variation than our year box plot



To better understand the patterns in our data, we used the `seasonal_decompose()` function to separate the data into trend, seasonal, and residual components. This again is difficult to see if we have seasonality, where there are parts where you can see some fluctuation but nothing extremely apparent.

Exogenous Variable Analysis

Analyzing exogenous predictors, feature selection.

To determine the most relevant exogenous predictors for our final forecasting model, we employed several analytical methods to assess correlation, collinearity, and feature importance. The process began with a check for missing values across all candidate predictors. Missing data was present in nearly all exogenous variables, so we addressed this by applying time-based interpolation followed by backward fill to handle any remaining gaps at the start of the series. However, two predictors contained no usable data across the entire time range and were therefore removed from consideration.

Next, we evaluated the correlation between each predictor and the target variable (`price_actual`). Predictors with a correlation coefficient below 0.2 were deemed insufficiently related to the target and were eliminated at this stage, reducing the number of candidate features by about half.

To further refine the selection, we applied Lasso Regression, which penalizes less informative predictors by shrinking their coefficients toward zero. This allowed us to isolate the variables contributing most strongly to the variability in energy prices. Through this method, we identified the following predictors as the most impactful: **generation_fossil_gas** (electricity generation from natural gas), **generation_fossil_hard_coal** (generation from hard coal), **generation_hydro_pumped_storage_consumption** (energy consumption used for hydro storage pumping), **generation_other_renewable** (generation from other renewable sources not explicitly categorized), and **total_load_forecast** (forecasted total electricity demand).

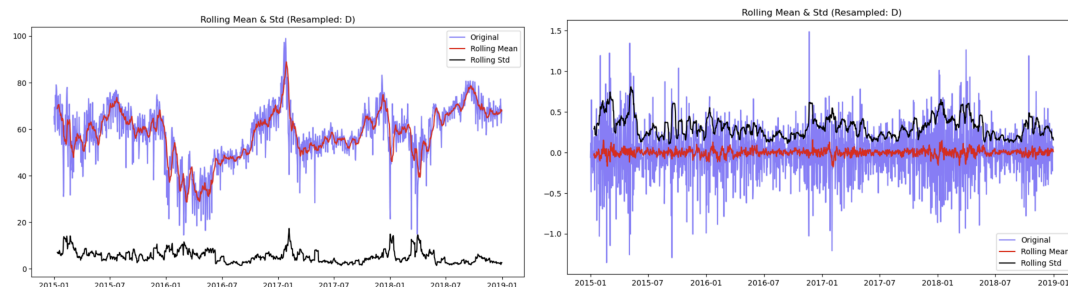
Finally, we assessed the selected features for multicollinearity using a correlation matrix. While some degree of multicollinearity was present among the remaining predictors, the correlation values did not exceed a level that would require additional variable removal. The highest correlation between predictors observed was between `total_load_forecast` and `generation_fossil_gas` (.54). This makes sense, as demand will naturally be correlated with generation.

Assumptions

Frequency, Stationarity

Before modeling, we needed to make sure the data met key time series assumptions regarding consistent frequency and stationarity. After setting "time" as the dataframe index, we confirmed that the dataset maintained a consistent hourly interval across all observations, with no missing timestamps. Having a fully consistent frequency is crucial for time series forecasting models, and since no irregularities were found, we were able to proceed without any resampling or adjustment.

Next, we checked for stationarity by first visually examining the `price_actual` variable over the four-year period. The first graph below shows the original price data (in blue), along with the rolling mean (red) and rolling standard deviation (black), resampled daily. The rolling mean shifts noticeably over time, and the standard deviation varies as well, suggesting strong non-stationary behavior with clear trends and fluctuations.



To address this, we'll apply first-order differencing in our models later. The results of this are shown in the second graph. After differencing, the rolling mean becomes much flatter, and the rolling standard deviation stabilizes, indicating that differencing helps remove trends and brings the series closer to stationarity.

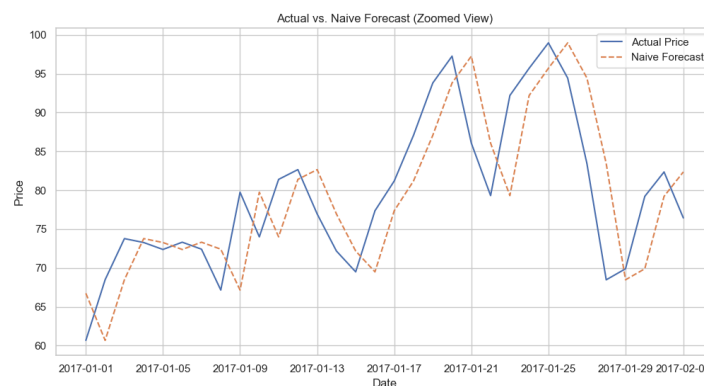
Furthermore, to formally verify stationarity, we performed the Augmented Dickey-Fuller (ADF) test. The results showed a test statistic higher than the 5% critical value and a p-value above 0.05, meaning we failed to reject the null hypothesis of a unit root and confirmed that the original price_actual series was non-stationary.

MODELING

The following section will outline the implementation and diagnostics of the following models: Naive Seasonal Model, ARIMA, SARIMAX, and XGBoost Stacking. A pyramid approach was used, starting with lower complexity models, then transitioning to more computationally involved models. Stacking was used to further improve predictions by combining the strengths of each individual model, allowing the ensemble to leverage different patterns captured by each approach. To assess our models' accuracy, we used the same measurements throughout. All models can be assessed using RMSE, MAE and MAPE. RMSE is an error metric that evaluates the error in the same scale as the dataset (Suleman, 2024). It highlights large errors in the prediction, which gives us a good look at whether our models can consistently give good predictions of prices. MAE is the second error metric that looks at the absolute difference of each prediction with the actual (Suleman, 2024). This is used to get a good basic summary of the average error, but does not give an in depth look at the data. MAPE is the final error metric we will use, which gets our average error into a percentage (Suleman, 2024). This is easiest to interpret, but can be sensitive to small errors.

Naive Seasonal Model

The first model chosen is Naive Seasonal Model, which we will use as our baseline model. Naive Seasonal Model takes a look at the seasonality of the models and uses the past season's fluctuations to predict the next cycle (Hyndman & Athanasopoulos, 2018a). This looks like making a prediction by shifting the data by a season's time, for our example we will shift by 24 hours. This means we are matching up each past hour of the day, so predicting 2:00 of today with the 2:00 of yesterday. This is a good baseline because it's a very simple prediction which makes it very efficient. With its simplicity, the model sacrifices in its accuracy of results.



Arima

Arima (or Auto Regressive Integrated Moving Average) is a common model used for Time Series, in which we combine three features: Auto-regression, differencing and moving average. Auto Regression is a linear combination of past variables used to predict the next variable (Hyndman & Athanasopoulos, 2018b). Differencing is used to fix stationarity issues. Moving average works similarly to AR, but instead of using past variables, it uses past errors (Hyndman & Athanasopoulos, 2018b). We found that our data was not stationary and we were able to fix that within the Arima model pretty easily,

with being able to use an integration (d) and moving average set (q) for 1. The more computationally expensive but important part of the arima for our project is the Autoregression (p) in which we choose how far back we want to read to get our future prediction. Using a higher value of 5 costs more but will let us look back at more data for each prediction we make, which lowers our efficiency but increases our accuracy.

At this point, it should be noted that we had to drastically reduce the number of observations considered for each model, as the computational power needed to make predictions for over 30,000 data points was simply too much for us to run in a reasonable amount of time. Because of this, we reduced the number of observations to the last 50 days of data, or the last 1,200 hours. Using a train-test-split of 90/10, we ended up generating 120 predictions total for each model. Any figures or diagnostics included are only in reference to this set, not the entirety of the data.

Sarimax

After ARIMA, we extended our modeling by incorporating SARIMAX which stands for Seasonal Auto Regressive Integrated Moving Average with eXogenous). SARIMAX is a time series model that builds upon ARIMA by adding two important features. This adds a seasonal PACF, where there is now a p, d, and q in a seasonal pattern as well (GeeksforGeeks, 2023). This also adds other variables in our model, where we are able to take into consideration how the other models can make trends from how the other variables move. In our project, the seasonal part was important because electricity prices tend to repeat certain patterns every week, month, or season depending on the weather and energy demand.

After running our SARIMAX model, we found that it gave us a better performance compared to our earlier ARIMA model. In SARIMAX, we used $(p,d,q) = (1,1,1)$ for the parameters and added seasonal parameters $(P,D,Q,S) = (1,1,1,7)$ to account for weekly seasonality. The seasonal period of 7 days made sense given the patterns we observed in energy prices, where prices tended to show weekly fluctuations.

While SARIMAX helped improve our predictions, there are some trade-offs between accuracy and efficiency. Since SARIMAX has to model both regular trends and seasonal patterns, it is more complicated and takes longer to train compared to simpler models like Naive Seasonal model or ARIMA. This means it can be slower and use more computing power. In our project, the dataset was small enough that the extra time wasn't a big problem, and the better predictions we got made it worth it. Overall, SARIMAX was a strong choice because it gave us more accurate results, and we had the time and resources to handle its extra complexity.

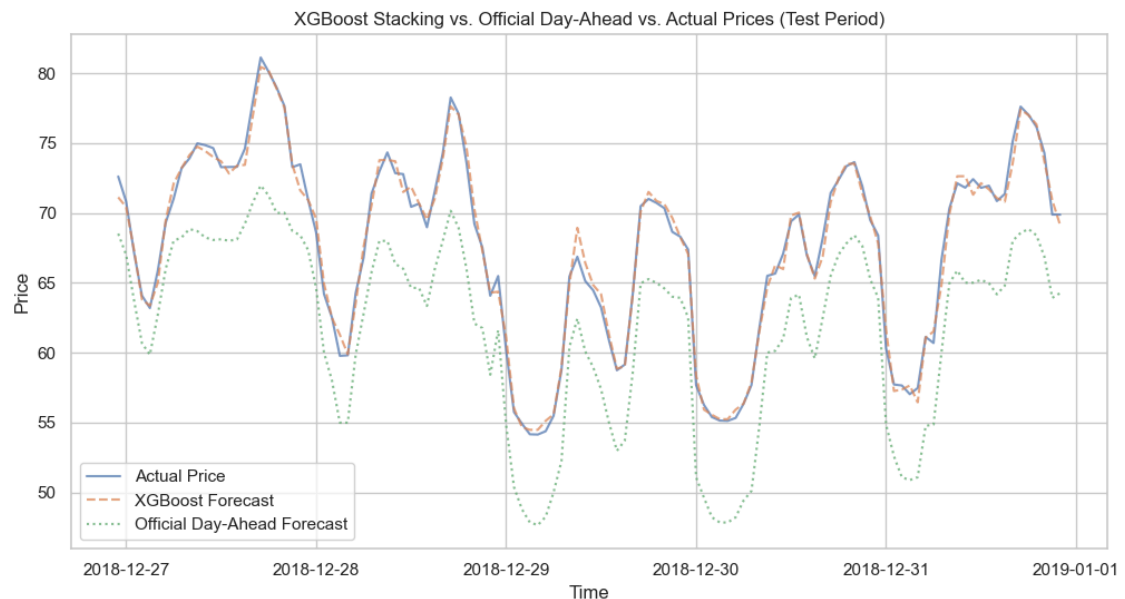
Ensemble Method

To enhance forecast accuracy beyond individual models, we implemented a stacking ensemble approach using XGBoost as the meta-learner. The model takes as input the predictions from the three base models: ARIMA, SARIMAX, and seasonal naive forecast. These outputs become features for the XGBoost regressor, where the model learns how to better weight and combine these individual forecasts. Unlike traditional time series models, which only look at past values of the target variable, the stacking approach leverages the strengths of each forecasting method, learning from their collective patterns to improve prediction accuracy.

RESULTS AND CONCLUSION

In this project we explored three different time series models to accurately predict electricity prices. The Naive Seasonal model was used as the baseline model to represent our dataset due to its simplicity as well as its low computational demand. Despite this efficiency, the accuracy our Naive model produced was limited. With an MAE value of 5.79, RMSE value of 8.31, and a MAPE of 11.75% we can conclude that our model would struggle with predicting future values, particularly sudden surges or drops in electricity prices that require more dynamic modeling. Since the RMSE is noticeably larger than the MAE, we can conclude that this model has some significantly large error predictors within the errors. In

comparison, our ARIMA model achieved a higher level of accuracy in forecasting prices. With an MAE value of 1.71, an RMSE value of 2.23 and a MAPE of 2.59%, we can confidently say that our ARIMA model could actively capture the short term price fluctuations in energy prices and properly structure future predictions. We can see a smaller difference between RMSE and MAE meaning our errors are getting better at consistency of error as well. However, there may still be shortcomings with this model that could be further investigated as longer term changes or patterns may have been missed or inaccurately interpreted. Our SARIMAX model produced an MAE of 1.43, RMSE of 1.90 and MAPE of 2.14% indicating that including seasonality and exogenous predictors into our forecasting process significantly improved the accuracy of our predictions compared to the previous model. While there may still be some variance with our predicted prices in comparison to the actual results, the overall accuracy of this model was quite good. Finally, the results of the ensemble stacking method using XGBoost demonstrate a significant improvement over the official day-ahead forecast. While the provided day-ahead forecast produced a MAE of 5.82, RMSE of 5.96, and MAPE of 9.57%, the XGBoost stacking model achieved a much lower **MAE of 0.54, RMSE of 0.70, and MAPE of 0.80%**. You can see there is a minimal difference between MAE and RMSE which means the errors are not only low, but consistently low. These results suggest that the ensemble approach, which combines multiple forecasting models, is highly effective at capturing the underlying patterns in the data and substantially reduces prediction error compared to the benchmark forecast used in industry. Below, the performance of our final XGBoost model is compared with price_actual and price_day Ahead.



Throughout the process of our research and model development, we faced a number of potential limitations on the accuracy and application of our results. One major constraint we encountered was the size of our dataset. Containing over 35,000 observations and spanning over the course of several years, the dataset provided an extensive overview of electricity prices, actively capturing seasonal effects as well as price fluctuations. However, due to the sheer volume of data present, the computing predictions proved to be an issue. Thus, when training our model we had to reduce the set to only 1200 observations. While necessary for our computation, this reduction in data likely introduced an extra level of bias into our model and limited the range of patterns and prices to be interpreted.

Given more time and resources, an important next step to address these issues would rely on training a larger number and wider variety of data points. Expanding the range of values and prices would allow for more accurate predictions as well as a lower chance of facing overfitting issues caused by more short-term trends. Expanding the range of values and prices included in training would improve the model's ability to generalize to real-world scenarios and strengthen its overall predictive performance.

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